

HEALTHCARE TREATMENT EFFECTIVENESS

Executive Report | Simulated De-identified Data

PATIENT ENCOUNTERS	1,800
MEAN IMPROVEMENT	10.8 points
FAVOURABLE OUTCOMES	50.2%
30-DAY READMISSION	11.6%

Executive Summary

This report analyses 1,800 simulated patient encounters. Mean clinical improvement is 10.8 points, the favourable-outcome rate is 50.2%, and 30-day readmission is 11.6%.

Treatment comparisons are stratified because age, baseline severity, condition, and comorbidity can influence both treatment selection and outcomes. Findings are descriptive associations, not causal estimates or clinical recommendations.

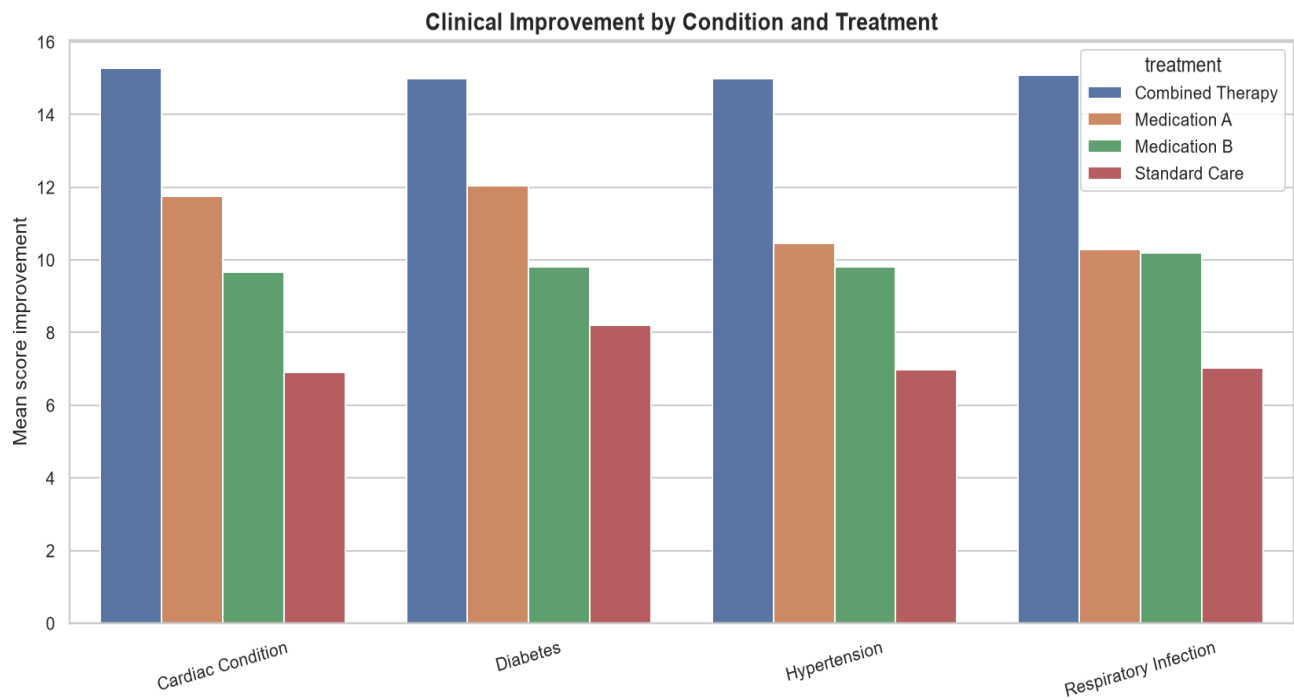
1. Methodology and Safety

Clinical improvement equals baseline severity score minus follow-up score; lower severity is better. A favourable outcome requires at least 10 points of improvement and no 30-day readmission. Data quality checks cover duplicates, ranges, missingness, and required fields.

The dataset is fully simulated and excludes direct identifiers. Real healthcare use would require validated clinical definitions, risk adjustment, clinician oversight, privacy governance, and fairness review.

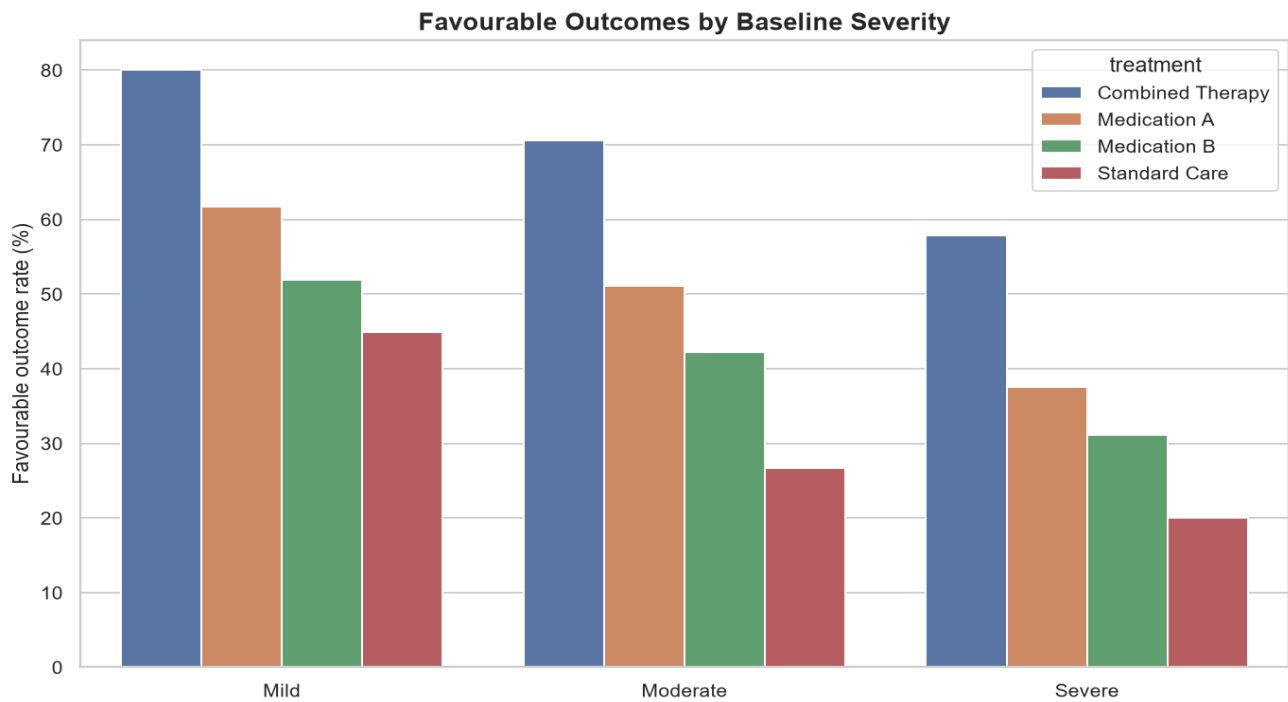
Metric	Value
Average age	55.6
Median improvement	10.7
Average stay	7.8 days
Median cost	INR 58,578
Average satisfaction	6.5 / 10
Cost-improvement correlation	0.040

2. Improvement by Condition and Treatment



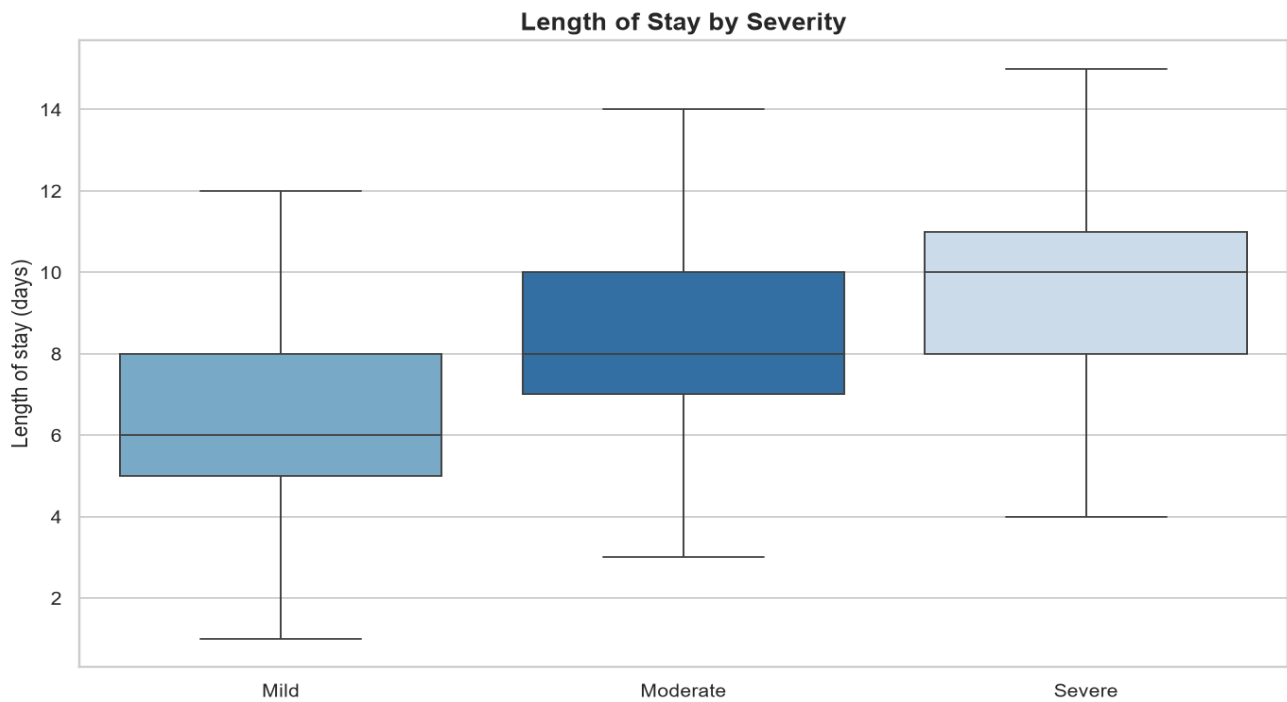
Condition-stratified improvement reduces, but does not eliminate, confounding from different patient mixes.

3. Favourable Outcomes by Severity



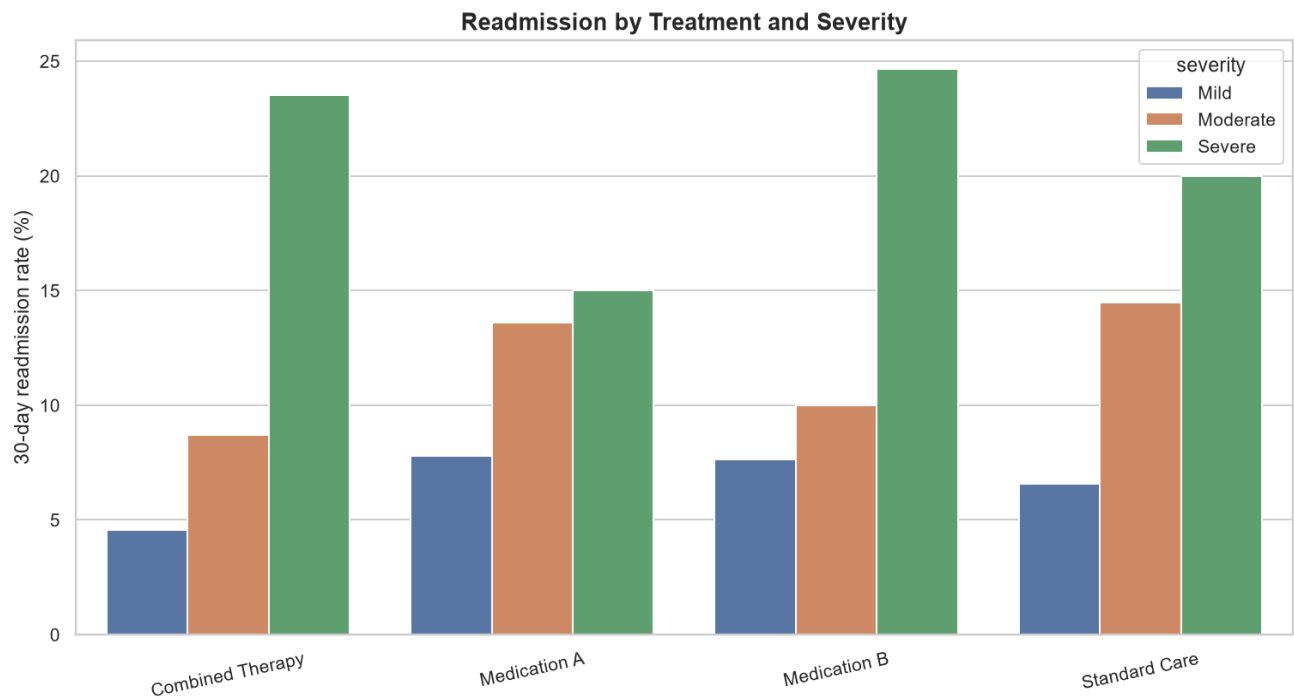
Outcome rates vary by severity, reinforcing the need to compare clinically similar groups.

4. Length of Stay



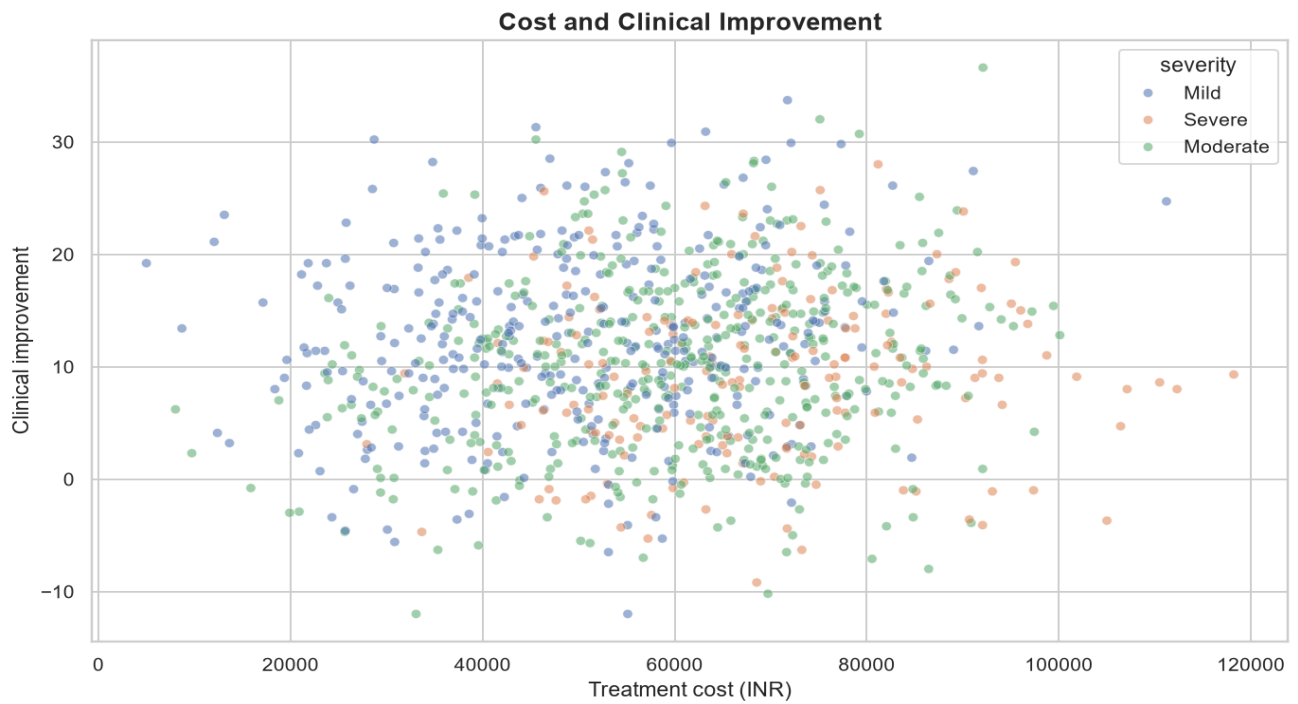
Stay distributions increase with severity and should be interpreted alongside clinical need, not as efficiency alone.

5. Readmission Patterns



Readmission differences may reflect severity and selection; investigate with risk-adjusted clinical review.

6. Cost and Improvement



Cost and improvement show wide variation. Higher cost cannot be assumed to cause better outcomes.

7. Recommendations and Limitations

Stratified review: Compare outcomes within the same condition and baseline severity.

Readmission review: Use structured clinical review for high-readmission groups; do not automate treatment changes.

Balanced scorecard: Monitor improvement, readmission, stay, cost, and satisfaction together.

Data governance: Audit definitions, missingness, coding consistency, privacy, and subgroup representation.

Clinical oversight: Require clinicians, ethics review, and risk adjustment before any real-world application.

Limitations

The data is simulated and observational. Treatment allocation is not randomised, residual confounding remains, and thresholds are portfolio definitions. Results must not be used for diagnosis, treatment selection, or patient-level decisions.